
From wholesale prices to consumer inflation

Layered pass-through of global oil shocks in India, 1983–2026

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Why oil-price pass-through matters for India

- India imports a large share of its crude oil.
- Crude is priced internationally, mainly in US dollars.
- Domestic oil pressure comes from Brent crude and INR/USD jointly.
- Oil moves through retail petrol, wholesale fuel, transport, and input costs.
- Headline CPI does not respond one-for-one with oil.



Objective and research question

Research question

How do global oil-price shocks transmit across India's wholesale, retail fuel, fuel-sensitive consumer, and headline consumer price layers, and where does this pass-through weaken?

Objective

- Estimate short-run oil-price pass-through across multiple Indian price layers.
- Identify the point at which the shock attenuates before reaching headline CPI.
- Compare strong and weak segments of the price system, layer by layer.

Contribution

- Treats oil pass-through as a layered transmission problem.
- Moves beyond a single oil-to-headline-CPI estimate.
- Covers five layers: WPI, WPI Fuel, retail petrol, CPI Fuel and Light, headline CPI.
- Tests attenuation directly on a common sample.

Data sources and variable purpose

Series	Source	Transformation	Purpose
Brent crude price	World Bank Pink Sheet (FRED POILBREUSDM)	Monthly log difference	Global oil shock
INR/USD exchange rate	FRED EXINUS	Monthly log difference	Exchange-rate component
Rupee oil price	Brent × INR/USD	Monthly log difference	Oil shock for WPI and headline CPI
Headline WPI	OEA, chained to 2011-12 base	Monthly log difference	Wholesale headline layer
WPI Fuel and Power	OEA, chained to 2011-12 base	Monthly log difference	Wholesale fuel-sensitive layer
PPAC Delhi retail petrol	PPAC ready reckoner	Monthly log difference	Direct retail fuel layer
CPI Fuel and Light	MoSPI harmonised series	Monthly log difference	Fuel-sensitive consumer bridge
Headline CPI	MoSPI / FRED CPI source	Monthly log difference	Consumer endpoint
IIP	MoSPI	Monthly log difference	Activity control

Short-run asymmetric ADL model

Estimating equation, by layer

$$\Delta y_t = \alpha + \sum \varphi_i \Delta y_{t-i} + \sum \beta_j^+ \Delta x_{t-j}^+ + \sum \beta_j^- \Delta x_{t-j}^- + \gamma' Z_t + \mu_m + \varepsilon_t$$

Model features

- Autoregressive — own lagged inflation captures persistence.
- Distributed lag — oil enters at lags 0–L.
- Asymmetric — positive and negative shocks enter separately.
- Short-run focus — monthly changes, no cointegration step.
- Controls: IIP for activity, month dummies for seasonality.

Reported quantities and inference

- CPT⁺** — sum of positive-shock coefficients across lags
- CPT[−]** — sum of negative-shock coefficients across lags
- p-values** — tests for CPT⁺, CPT[−] and CPT⁺ = CPT[−]
- Estimation** — OLS
- Inference** — Newey–West HAC standard errors

Main result: pass-through weakens across layers

Layer	Sample	N	CPT ⁺	p	CPT ⁻	p	Asym. p	Reading
WPI Fuel and Power	2010-04 to 2026-03	186	0.5205	<0.001	0.4195	<0.001	0.1642	Strong wholesale fuel
PPAC retail petrol	2004-08 to 2024-12	245	0.3459	<0.001	0.1912	0.0002	0.0999	Strong direct fuel
CPI Fuel and Light	2011-05 to 2024-12	164	0.1777	0.0021	0.1058	0.1741	0.4554	Bridge evidence
Headline WPI	1983-05 to 2026-03	515	0.0301	0.0240	0.0374	0.0012	0.6727	Modest but significant
Headline CPI	2004-08 to 2024-12	245	0.0213	0.1220	0.0006	0.9375	0.2408	Weak, not significant

Reading

Pass-through is strongest in WPI Fuel and Power and PPAC retail petrol, smaller in CPI Fuel and Light, modest in headline WPI, and weak and statistically insignificant in headline CPI. Diagnostics pass for each claim-bearing specification (BG, HAC-RESET, recursive CUSUM).

Wholesale results: WPI and WPI Fuel and Power

Headline WPI

Sample: 1983-05 to 2026-03 (N = 515)

$CPT^+ = 0.0301$ $p = 0.0240$

$CPT^- = 0.0374$ $p = 0.0012$

Modest but statistically significant.

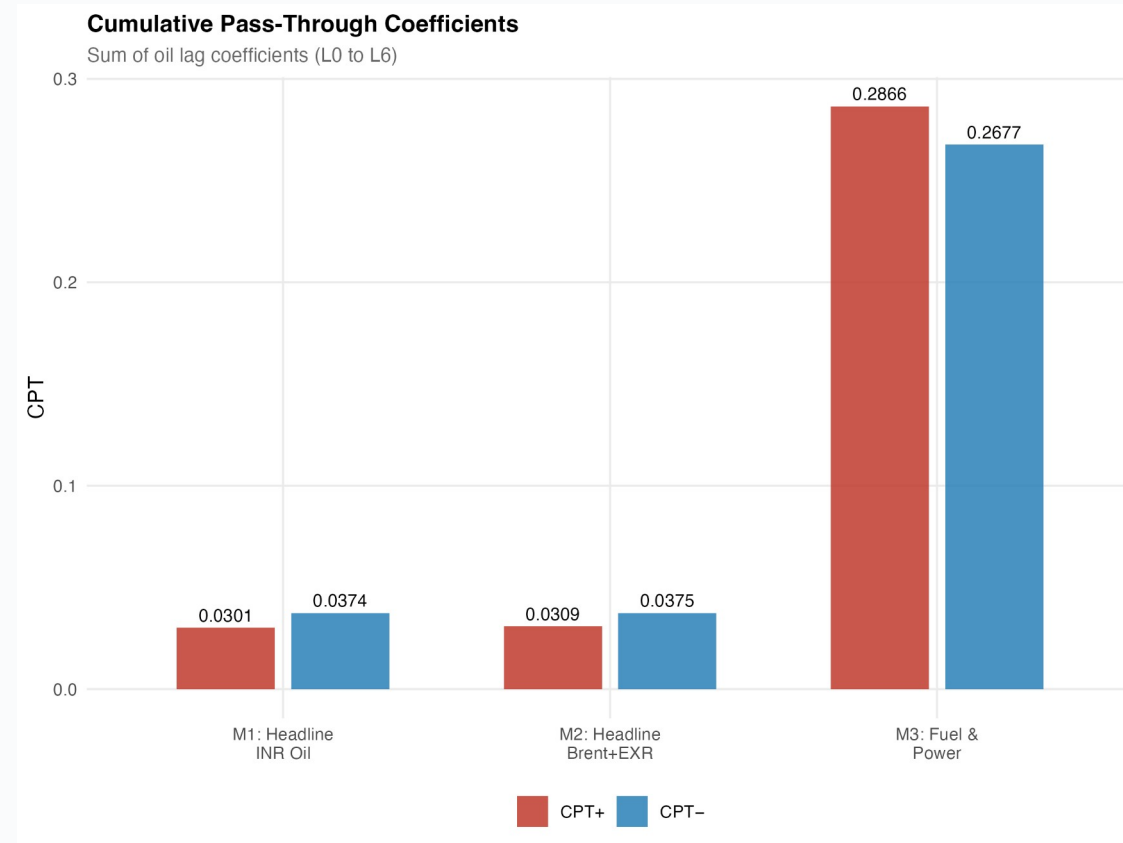
WPI Fuel and Power • preferred

Sample: 2010-04 to 2026-03, excl. Apr–Sep 2020 (N = 186)

$CPT^+ = 0.5205$ $p < 0.001$

$CPT^- = 0.4195$ $p < 0.001$

Strong wholesale fuel pass-through. Post-2010 sample, COVID months excluded.



Cumulative pass-through across WPI specifications

Consumer fuel layers: retail petrol and CPI Fuel and Light

PPAC retail petrol • shock = Brent

Sample: 2004-08 to 2024-12 (N = 245)

CPT⁺ = **0.3459** p < 0.001 | CPT⁻ = **0.1912** p = 0.0002

Asymmetry p = 0.0999 (marginal at 10%). Strong direct fuel pass-through.

CPI Fuel and Light • shock = PPAC retail petrol

Sample: 2011-05 to 2024-12 (N = 164)

CPT⁺ = **0.1777** p = 0.0021 | CPT⁻ = **0.1058** p = 0.1741

Bridge from retail fuel to the consumer fuel basket; effect smaller than upstream petrol.

Headline CPI: where the oil signal becomes weak

Headline CPI model

Sample: 2004-08 to 2024-12 (N = 245)

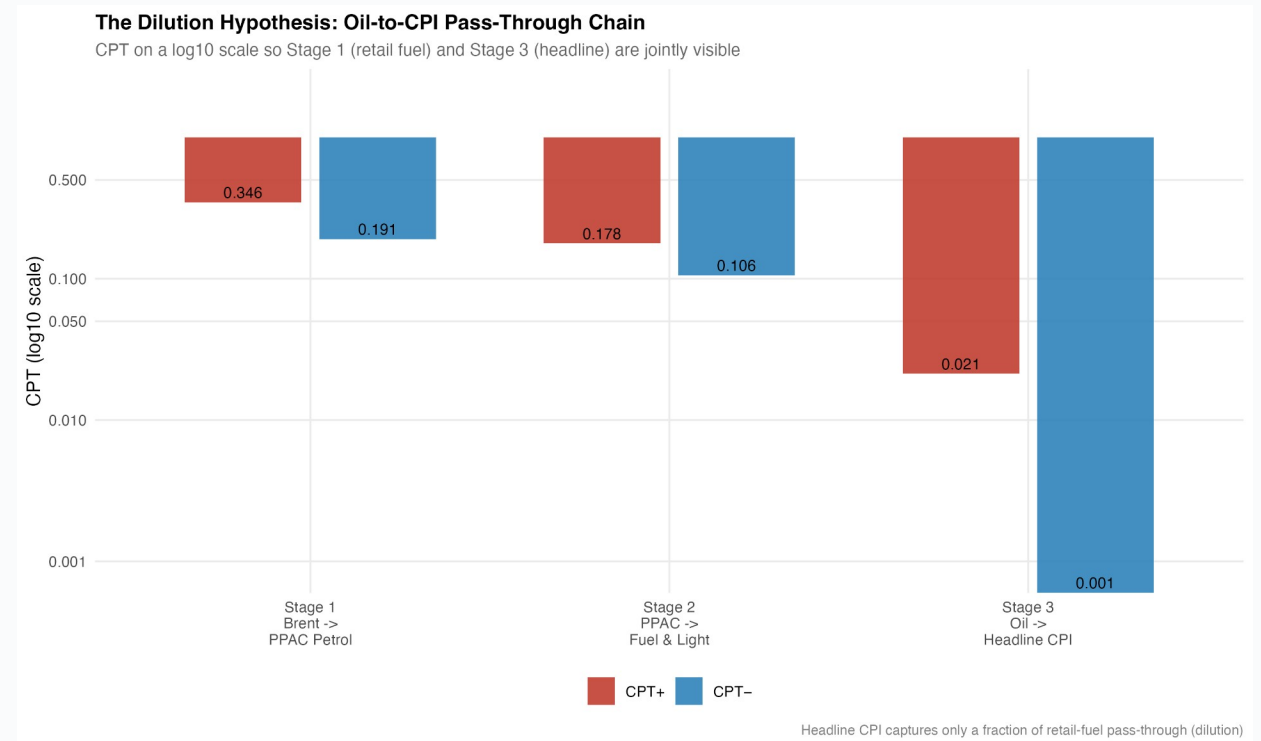
$CPT^+ = 0.0213$ $p = 0.1220$

$CPT^- = 0.0006$ $p = 0.9375$

Asymmetry $p = 0.2408$

Reading

- Positive CPT^+ has the expected sign but is not significant.
- Headline CPI is broad and food-heavy; fuel weight is limited.
- Taxes, margins, and policy can absorb part of the shock.
- This is attenuation, not absence.



Cumulative positive pass-through across the consumer-price chain

Attenuation across the consumer chain

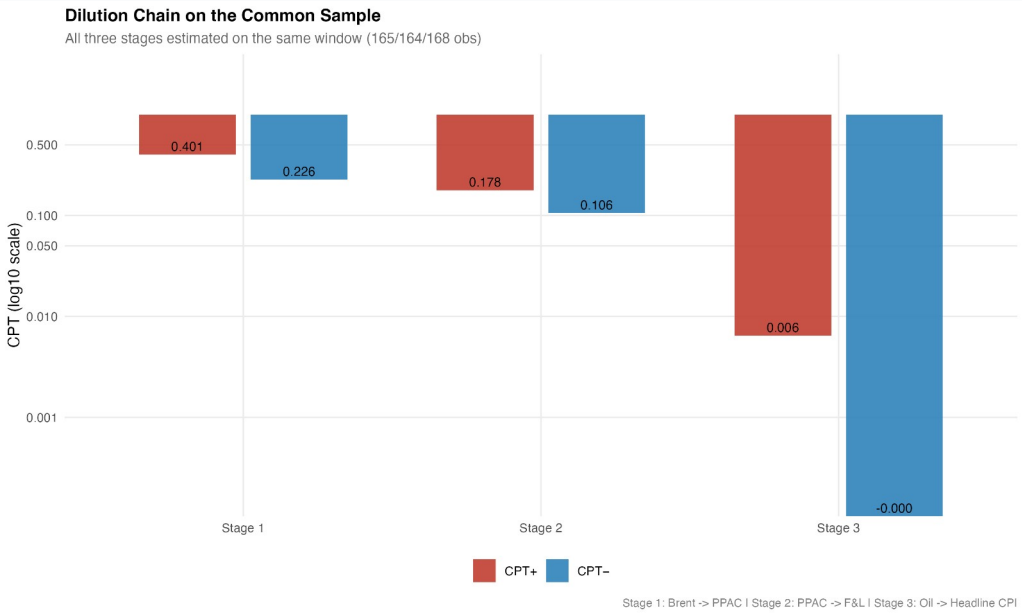
Stage	Relationship	CPT ⁺	p
Stage 1	Brent → PPAC retail petrol	0.4007	0.0001
Stage 2	PPAC petrol → CPI Fuel and Light	0.1777	0.0021
Stage 3	Rupee oil → Headline CPI	0.0064	0.6355

Pass-through shrinks across stages

Stage 1 vs Stage 3 ratio: $0.40 / 0.006 \approx 63\times$ smaller

Coefficient-equality test (Stage 1 = Stage 3): $F = 14.35$, $p = 0.0002$

Equality is rejected; the headline-CPI response is significantly smaller than the upstream retail-petrol response.



Limitations

Scope and inference

- Reduced-form short-run projections; not structural causal estimates.
- Layered map, not a single mechanical WPI-to-CPI chain.
- Granger checks signal predictive precedence, not deep causality.
- Asymmetry is tested but treated as suggestive.

Data and design

- Layers have different sample lengths because of data availability.
- CPI Fuel and Light has a shorter sample; treated as bridge evidence.
- PPAC retail petrol uses Delhi prices as a national retail-fuel proxy.
- The pre/post-2010 split is institutional context, not a clean experiment.

Conclusion: oil pass-through is layered and attenuated

Oil shocks matter in India, but their effect weakens across the price system.

Findings by layer

Strong — WPI Fuel and Power and PPAC retail petrol

Bridge — CPI Fuel and Light: smaller but significant

Modest — Headline WPI: small, statistically significant

Weak — Headline CPI: not statistically significant

Takeaways

- Attenuation pattern is visible at every step of the consumer chain.
- Stage 1 vs Stage 3 ratio: about 63× smaller; equality test rejected.
- Asymmetry is not the central result; only retail petrol is marginal.
- Fuel-only and headline-CPI views give different policy reads.

Attenuation, not absence, of oil-price transmission.

Thank you

Questions are welcome.



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